IDEA 1:

Smart Local Market Inventory & Sales System

Background (fictional but realistic)
A small neighborhood supermarket wants a simple database system to track products, suppliers, sales, and customers.

Purpose of the system
Replace manual records with a centralized database to improve inventory control and sales analysis.

Main entities

- Products
- Categories
- Suppliers
- Customers
- Sales
- Sales_Details
- Employees

Key functionalities

- Add and update product stock
- Track daily sales
- Automatically reduce inventory after each sale
- Identify low-stock items
- Generate sales reports by date or category

Market Inventory & Sales Management System

1. Stakeholders

- Market owner / shop proprietor
- Store manager
- Cashiers
- Suppliers
- Customers (indirect stakeholders)

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2. Users

- **Admin (Owner/Manager)** – full system control

- **Cashier** – sales entry and billing
- **Storekeeper** – inventory updates and stock checks

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3. Ownership

- Owned and controlled by the **market owner or business entity**
- Data belongs to the business
- Access rights defined by user roles (admin, cashier, storekeeper)

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4. Finance (Sponsorship & Funding)

- Self-funded by the market owner
- Ongoing costs:
  - System maintenance
  - Hardware (computers, barcode scanners)
  - Staff training

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5. Tools

- **Database:** MySQL / PostgreSQL
- **Frontend:** HTML, CSS, JavaScript (or MS Access forms)
- **Backend:** PHP / Python / Java
- **Modeling Tools:** Draw.io, Lucidchart (for ERD & DFD)
- **Development Tools:** VS Code, XAMPP

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6. Basic Functionalities

- Add, update, and delete product records
- Track stock levels automatically after each sale
- Record daily sales transactions
- Generate invoices/receipts
- Manage supplier information
- Generate reports (daily sales, low stock, top-selling products)

IDEA 2:

University Residence Visitors & Access Control System

(Adapted from apartment system to university hostels/residence halls)

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1. Stakeholders

- University administration
- Residence hall management
- Security department
- Students (residents)
- Visitors (indirect stakeholders)

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2. Users

- **System Administrator** – overall system management
- **Security Personnel** – visitor entry/exit logging
- **Residence Supervisor/Warden** – monitoring and reporting
- **Students (limited access)** – view visitor history or approve visits (optional)

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3. Ownership

- Owned by the **university**
- Visitor data governed by university data protection policies
- Centralized access managed by the IT department

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4. Finance (Sponsorship & Funding)

- Funded by the university
- Costs include:
    - System development
    - Hardware (PCs, scanners, ID readers)
    - Security staff training

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5. Tools

- **Database:** MySQL / Oracle / PostgreSQL
- **Frontend:** Web-based interface or desktop application
- **Backend:** Java / PHP / Python
- **Modeling Tools:** StarUML, Draw.io
- **Security Tools (conceptual):** ID verification, QR code scanning

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6. Basic Functionalities

- Register resident student details and room assignments
- Record visitor information (name, ID, contact)
- Log entry and exit times
- Link visitors to specific students and rooms
- Allow supervisors to view visitor reports
- Generate security logs and visit history reports